Visualising Emotion

Measuring agreement between a written feeling and an AI-generated drawing

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THE PROBLEM

Object prompts can be checked. Feelings cannot.

“a red car”

Is there a car? Is it red? The prompt names objects, so existing metrics can score it.

“tired but happy”

No object to find, no answer key, and the feeling is mixed. Existing metrics do not transfer.

Research question

To what extent does an automatic image-emotion reader agree with prompt-derived emotion labels for AI-generated monochrome drawings, and how does that agreement vary by visual content and by repeated prompts?

Sub-questions: (i) does agreement exceed a random-pairing baseline? (ii) does it differ for face-classified and scene-classified drawings? (iii) how stable is the image reading when the same prompt is drawn again? (iv) does describe-then-extract agree more than reading emotion from the image directly?

The question is deliberately narrower than “does the drawing convey the feeling” — see the closing slide.

LITERATURE

Four established areas, and the space between them

Emotion in text	Well established. GoEmotions supplies a 27-category taxonomy and a classifier; but short first-person self-report is the hardest register.
Emotional image generation	Well established. Systems generate images to carry a chosen emotion, and colour is reported as the principal carrier of affect.
Text-to-image evaluation	Well established for OBJECTS. CLIPScore and VQAScore measure whether the picture contains what the prompt named.
Reading emotion from an image	Weak. Vision-language models are positively biased and order-sensitive readers of affect.

The gap: no study has measured whether the emotion in a person's own unedited self-report is recoverable from the drawing generated from it — in monochrome line art, where the reported colour channel is absent.

97 papers screened · 23 excluded · 74 retained across twelve themes. Colour is cited as motivation only — this study has no colour-control condition.

CONTRIBUTION

Similarity, not matching

“Asking whether the image emotion matches the text — right or wrong — is too simplistic. Emotion is subjective, there is no ground truth, humans disagree and models are noisy. Ask how similar the two emotions are.”

— Supervisory direction, 30 July 2026

Conventional

One-to-one match. The predicted label is right or wrong; accuracy is reported against an assumed ground truth.

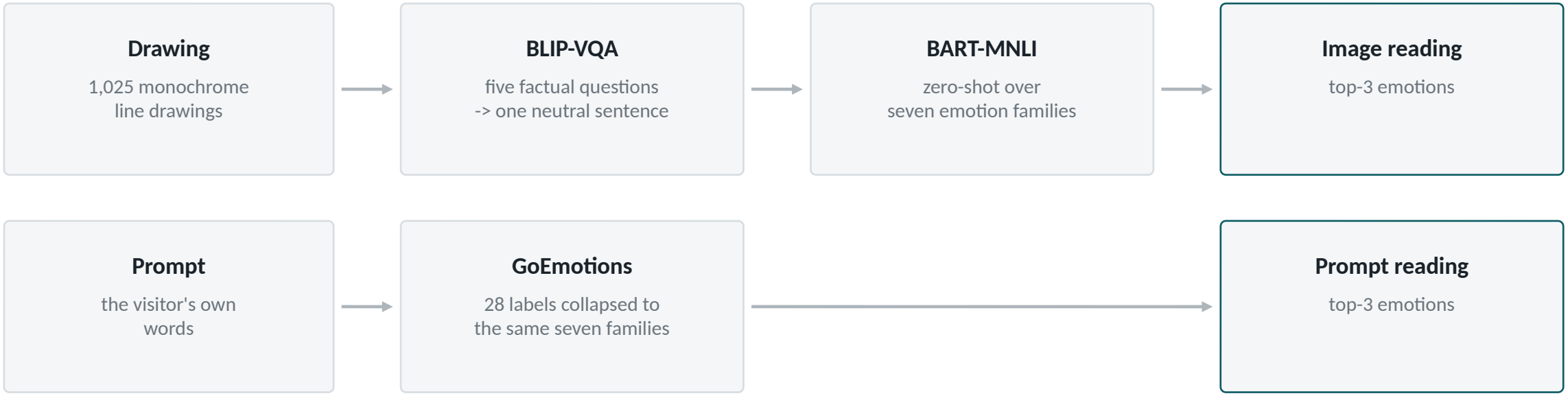
This project

A four-level similarity ladder over emotion families. “Happy” and “excited” are counted as close, not as a miss — and every level is referenced to chance.

This is the novel element: the field reports one-to-one accuracy; this study reports graded, chance-referenced similarity.

METHODOLOGY

Describe the image, then read the feeling from the description



Compare → four-level similarity ladder, every level against a 300-shuffle random-pairing null

- valence-arousal distance
- valence-sign agreement
- emotion-family agreement
- top-3 overlap

The image model is asked only factual questions. Caveat stated in the write-up: "what is the facial expression?" often returns a feeling word, so the separation is weaker than it appears.

DATA

1,025 real self-reports, and what it took to use them

1,012
analysed records
(13 pre-festival tests excluded)

607 / 405
prompts carrying a feeling
versus not — the specificity set

591 / 421
face-classified versus
scene-classified drawings

87
prompts drawn more than once,
covering 336 records

Data engineering

- Three incompatible filename schemes in one folder — a naive loader resolves under 20%.
- C2PA signing timestamps recovered from inside each PNG identified 13 pre-festival developer tests.
- Ran to completion on all 1,025 images with no unhandled errors — a robustness statement, not an accuracy one.

Data pre-processing

- Eight records contained names of private individuals; these were replaced with a placeholder.
- Detection combined a 7,944-name gazetteer, a 236,000-word dictionary and naming phrases — 77% of the corpus is lower case, so capitalisation alone would miss most names.
- Verified inert: re-running the prompt reader changed no emotion label.

EVALUATION DESIGN

Two checks that decide whether any number means anything

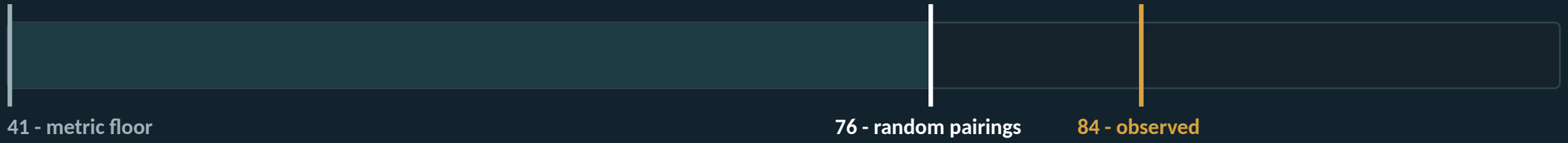
1 · A random-pairing null

Within each subset the image side is shuffled and re-paired against the untouched prompts, 300 times, with a fixed seed. Every figure I report is quoted against its own null, so a result counts only insofar as it exceeds chance.

2 · A specificity check

405 prompts carry no feeling — “pigs”, “mickey mouse”. On these the valence–arousal metric returns 76.6% against a null of 76.6%, z = +0.3. The pipeline reports agreement only where the prompt reader finds an emotion.

And the scale does not start at zero



Consequence: a valence–arousal score of 71 reads as a pass but is below chance. Reporting therefore leads with valence-sign and family agreement, which the metric cannot substitute for.

RESULTS

Agreement is above chance, and strongest for face-classified drawings

Subset	Valence-sign agreement	Emotion-family agreement	Verdict
Emotional prompts (n=607)	64.0% vs null 45.6% z +11.1	53.1% vs null 38.8% z +10.3	Well above chance
Face-classified (n=402)	73.9% vs null 53.5% z +10.4	61.9% vs null 45.8% z +10.3	Strongest
Scene-classified (n=205)	42.9% vs null 29.6% z +5.8	35.6% vs null 26.2% z +4.4	Above chance, weaker
Non-emotional (n=405)	— degenerate by definition	— degenerate by definition	Specificity check

Repeated prompts

Given the same prompt text, the image reader returns the same emotion for 68.8% of drawings within a group, against a permutation null of 59.1% (z +4.9). The generator is not deterministic in what the reader recovers.

Method comparison

Holding the prompt reader constant and varying only the image reader: describe-then-extract 49.3% family agreement against direct-CLIP 25.3%. Note the prompt reader here is BART-MNLI, not the GoEmotions reader used above.

Robustness

Re-run on the 763 distinct prompts, the overall and face results hold (z +9 to +11). The scene result weakens substantially, to 37.3% (z +3.5).

No formal significance test of the face–scene difference has been run; both rates are reported against their own nulls instead.

DEMONSTRATION

The pipeline, running live

1

Choose a drawing

from the festival dataset, or upload an image

2

Watch it described

the five factual answers, then the emotion read from them

3

Compare

both readings, and the score against the chance line

Every figure the interface shows is referenced to the measured chance line, and nothing a visitor types is stored.

CONCLUSION

What this establishes, and what it does not

Established

- Prompt-reader and image-reader agreement is well above chance on every metric.
- Agreement is substantially higher for face-classified than scene-classified drawings.
- The same prompt does not reliably yield the same recovered emotion (69% vs 59% null).
- Describe-then-extract agrees more than reading emotion from the image directly.

Not established

- What a human viewer perceives — both readers are automatic, and they disagree with each other.
- That the face–scene gap is statistically significant; no formal test has been run.
- Any effect of colour — the corpus is monochrome but there is no colour condition.
- Any positive-versus-negative direction; that claim was withdrawn.

Future work: human validation of both readers · sketch-trained describers · a formal test and confidence intervals for the face–scene difference

The contribution is a chance-referenced similarity measure for emotional agreement, not a new classifier.

Thank you — questions welcome